

MANAR: Supervised Autonomy, Deterministic Control, and Greedy Search Route Optimization for Aerial Search-and-Rescue

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Abstract—Aerial Search-and-Rescue (SAR) operations require systematic area coverage, dependable navigation, and robust operational safety considerations. This paper presents the architecture, mathematical design, and functional prototype of MANAR (Multi-sensor Autonomous Navigation and Aerial Rescue), a supervised-autonomy SAR system. We draw a clear distinction across three system maturity tiers: the currently implemented C++ functional prototype, the selected closed software architecture, and planned future perception extensions. The core technical contribution focuses on MANAR’s deterministic control flow, Haversine geographic navigation, and an $O(n^2)$ greedy nearest-neighbor search route optimizer. Crucially, the system enforces operator override privilege: automated route optimization assists the operator without overriding manually entered sector sequences when tactical domain knowledge dictates search ordering. We present parameterized mathematical models for mode-dependent velocity profiles, simulated battery failsafe progression, and systematic lawnmower grid search. Functional prototype verification confirms deterministic control behavior, search-plan locking, and Return-to-Launch (RTL) navigation preemption.

Index Terms—Search-and-Rescue UAVs, Deterministic Control, Supervised Autonomy, Greedy Route Optimization, Haversine Navigation, Autonomous Systems.

I. INTRODUCTION

Search-and-Rescue (SAR) missions represent a critical operational domain for unmanned aerial systems [1]. When survivors are stranded in vast, inaccessible terrain, emergency response windows are severely constrained by environmental conditions. Unmanned Aerial Vehicles (UAVs) provide rapid deployment, flexible sensor mobility, and elevated coverage.

MANAR (Multi-sensor Autonomous Navigation and Aerial Rescue) is a supervised-autonomy SAR system designed to balance human oversight against automated execution [2]. The term *Autonomous* refers specifically to deterministic navigation capabilities under supervisory human authority. Fully manual teleoperation imposes heavy cognitive workloads on operators, while unconstrained autonomous systems introduce failure risks during sensor dropouts or communication loss. In MANAR, human operators retain supervisory authority over mission configuration and target verification, while a C++ control core executes deterministic navigation, route optimization, and safety monitoring.

TABLE I
MANAR CAPABILITY IMPLEMENTATION STATUS

Capability	Status
Deterministic C++ control core	✓ Implemented
Haversine navigation geometry	✓ Implemented
Multi-location sector sequencing	✓ Implemented
Greedy route optimization ($O(n^2)$)	✓ Implemented
Operator optimization toggle (Y/N)	✓ Implemented
Lawnmower search grid logic	✓ Implemented
Payload component state controls	✓ Implemented
Structured event logging	✓ Implemented
Visual candidate detector (YOLO11n + ORT)	✓ Implemented
Reference detector baseline (D-FINE-N)	✓ Implemented
Empirical 26-video SAR benchmark	✓ Implemented
React / TypeScript Web GUI	Selected
WebSocket + JSON IPC transport	Selected
Authoritative C++ state ownership	Selected
Candidate tracking & Kalman filtering	Planned
Multimodal evidence fusion (Mamba SSM)	Planned
PX4 SITL / HITL simulation	Planned
Physical flight deployment	Planned

A. Three-Tier Maturity Framework

To maintain engineering rigor, MANAR’s capabilities are categorized across three distinct maturity tiers, summarized in Table I:

- 1) **Implemented Functional Prototype:** Fully compiled C++ control logic, double-precision Haversine navigation, $O(n^2)$ greedy route optimization with operator override, lawnmower sector sweeps, discrete battery simulation, structured event logging, and an ONNX Runtime C++ visual human detector module (YOLO11n) benchmarked against a D-FINE-N reference baseline.
- 2) **Selected Closed Architecture:** Authoritative state ownership in C++, React/TypeScript web interface over WebSocket + JSON, and on-demand Return-to-Launch (RTL) navigation semantics.
- 3) **Planned Future Work:** Multi-frame candidate tracking (Kalman / ByteTrack), temporal multi-sensor evidence fusion (Mamba SSM), PX4 SITL/HITL flight integration, and field deployment.

B. Key Contributions

The main contributions of this work are:

- Formulation and verification of an $O(n^2)$ greedy nearest-neighbor route optimizer using Haversine geodetics [3] with operator override privilege.
- Integration and empirical validation of an ONNX Runtime C++ visual human detector module (YOLO11n) achieving 29.2 ms latency with zero false positives on clutter across 26 SAR video datasets.
- Specification of a dual-regime adaptive telemetry rate policy and analytical radio power consumption model across transmission frequencies.
- Formalization of parameterized motion, two-level RF attention, and simulated battery failsafe progression models.
- Functional verification of search plan locking, sequential sector progression, and Return-to-Launch navigation pre-emption.

II. SYSTEM REQUIREMENTS

MANAR is engineered to satisfy operational and safety constraints for aerial search-and-rescue.

A. Operational Envelope and Performance Targets

The system targets an operational flight envelope tailored for rugged wilderness and desert environments:

- **Search Endurance (SYS-G-02):** At least 1.0 hour of continuous active search under a nominal payload and representative sensor duty cycle.
- **High-Altitude Operational Ceiling (SYS-G-01):** Takeoff, climb, and stable search up to 2000.0 m AMSL with designated payload.
- **Speed Presets and Limits:** Four discrete velocity presets (v_Q, v_A, v_I, v_H) bounded by software velocity clamp $v_{\max}$ and altitude ceiling $H_{\max}$. Takeoff climb target is specified by H_{launch} .

B. Terrain-Adaptive Search Strategy

Search path generation adapts to local terrain geometry:

- **Open / Flat Terrain:** Parallel-track lawnmower sweeps parameterized by sector width W , height H , and row spacing Δy .
- **Mountainous / Steep Terrain:** Terrain-aware contour-following paths maintaining nominal sensor standoff distance.

C. Supervised-Autonomy Responsibility Allocation

Table II delineates the functional boundaries between the human operator, deterministic control logic, and machine learning pipelines.

TABLE II
SUPERVISED-AUTONOMY RESPONSIBILITY ALLOCATION

Layer	Core Responsibilities
Human Operator	Mission definition, geofencing, takeoff approval, real-time RTL/waypoint override, alert review, final <i>Probable Rescuee</i> determination, guidance authorization.
Deterministic Core	Flight dynamics, geodetic navigation, safety clamps, candidate approach trajectories, failsafes, sensor synchronization, DSP filtering.
Machine Learning	Computer vision candidate detection, acoustic classification, radar micro-motion classification, Mamba hover verification alert gating. (ML never directly commands actuators).

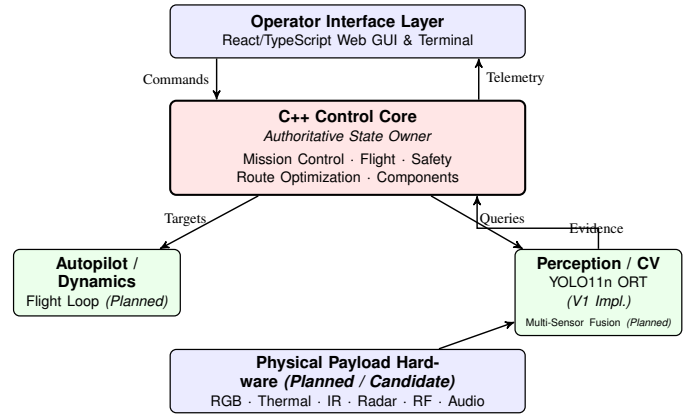


Fig. 1. MANAR system architecture. The C++ Control Core is the authoritative state owner governing navigation, safety, and mission progression.

D. Deterministic Contingency and Failsafe Rules

The flight system enforces deterministic contingency responses:

- **Command-Link Loss (SYS-R-01):** Upon link interruption, the aircraft maintains a temporary hold or verified safe route. If reconnection fails within a configurable timeout, autonomous RTL is engaged.
- **Energy Failsafe (SYS-R-02):** Sufficient return energy is continuously reserved. When return capability is compromised, RTL is triggered; if return is unachievable, an emergency landing sequence executes while preserving coordinates.
- **GNSS Degradation:** Transitions to attitude hold and local clearance sensing without making unverified absolute radar localization claims.
- **Environmental Limits:** Dangerous wind, icing, or propulsion anomalies automatically trigger mission abort or RTL.

III. SYSTEM ARCHITECTURE

MANAR's architecture establishes clear functional boundaries across operational layers. Fig. 1 illustrates core components and data flows.

A. Selected State Ownership Architecture

MANAR's selected final architecture establishes that the C++ Control Core owns authoritative runtime state. Under this design, state modifications requested by operators or triggered by safety rules are evaluated and committed exclusively by the C++ control layer (`control.exe`). Subsystem modules update internal state variables and report telemetry to the core rather than directly mutating shared state structures.

Implementation Status Note: In the current software prototype, several subsystem functions still persist runtime data directly to disk via `saveRuntime()`. These calls are being systematically refactored toward the centralized state-ownership model.

B. Communication & IPC Architecture

The communication architecture distinguishes between current prototype IPC and the selected final interface design:

- **Current Prototype Data Flow:** File-backed JSON contracts under `runtime/` handle prototype communication and state persistence: `commands.json` serves as the command/request transport from the terminal to the control application, `runtime.json` provides the live prototype status representation, and `search_locations.json` stores the persisted locked search-plan representation written by `control.cpp` upon processing location entry commands.
- **Selected Final Architecture:** Real-time WebSocket + JSON protocol between the React/TypeScript GUI and C++ control core. In this target model, `runtime.json` is retained as an on-demand snapshot export rather than the live transport.

IV. MISSION ARCHITECTURE AND CONTROL FLOW

MANAR coordinates flight execution through a modular navigation hierarchy and deterministic mission logic.

A. Navigation vs. Payload Sensor Layering

To safeguard flight stability, MANAR decouples flight-critical navigation sensors from mission payload instrumentation:

- **Navigation Sensor Suite:** Dual GNSS/GPS, IMU, 3-axis magnetometer, barometric altimeter, downward radar altimeter (terrain-relative clearance), and forward/lateral obstacle radar interfacing with a Pixhawk-class autopilot.
- **Payload Sensor Suite:** Dedicated 24 GHz rescue FMCW radar, thermal LWIR, RGB, low-light IR cameras, digital MEMS acoustic array, and passive RF software-defined radio.

B. Conceptual Mission Lifecycle

Fig. 2 depicts the complete 10-stage conceptual mission lifecycle from deployment to recovery.

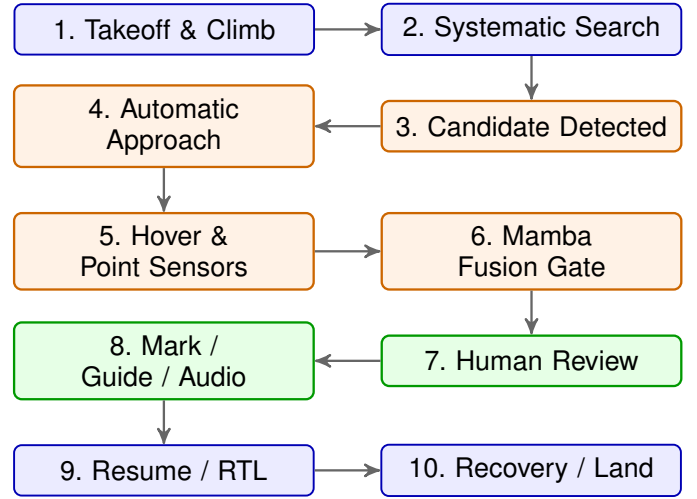


Fig. 2. Conceptual 10-stage search, verification, and rescue lifecycle.

C. Candidate Approach and Hover Verification Sequence

Upon wide-area candidate detection, the system executes a deterministic 9-step verification sequence:

- 1) **Evidence Registration:** Log candidate coordinates and sensor detections.
- 2) **Constraint Validation:** Check geofence, terrain clearance, wind, and battery reserve.
- 3) **Velocity Deceleration:** Automatically reduce horizontal speed to v_I .
- 4) **Safe Descent:** Descend to nominal inspection altitude H_{inspect} .
- 5) **Positioning Orbit:** Execute bounded inspection orbit if required.
- 6) **Gimbal Aiming:** Orient imaging, radar, and acoustic sensors toward target.
- 7) **Hover Stabilization:** Transition to stationary hover ($v_H = 0$).
- 8) **Synchronized Verification:** Execute acoustic call-and-listen cycles while evaluating Mamba temporal fusion.
- 9) **Alert Escalation:** Trigger ground station alert upon meeting fusion threshold for human review.

D. Prototype Control Flow & RTL Semantics

In the current C++ prototype, multi-location search plans execute sequentially via `mission.cpp`. When sector k completes ($1 \leq k \leq N$), the control loop advances $k \leftarrow k + 1$ and re-initializes sweep counters. Upon completion of all N sectors or upon failsafe trigger, Return-to-Launch (`activateRTL()`) is commanded, resetting navigation destination to home coordinates (`isdestinationhome = true`).

V. MATHEMATICAL MODEL

MANAR incorporates geographic navigation geometry, flight dynamics rules, and battery simulation models to govern aircraft behavior.

A. Geographic Distance (Haversine Formulation)

To evaluate spatial relationships over Earth's spherical surface without relying on planar approximations, MANAR calculates geographic distance using the Haversine formula.

Let two points on the Earth's surface be defined by their geodetic coordinates:

$$P_1 = (\phi_1, \lambda_1), \quad P_2 = (\phi_2, \lambda_2) \quad (1)$$

where ϕ represents latitude and λ represents longitude in radians.

The angular differences are:

$$\Delta\phi = \phi_2 - \phi_1, \quad \Delta\lambda = \lambda_2 - \lambda_1 \quad (2)$$

The square of half the chord length between the points is given by:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right) \quad (3)$$

The angular distance in radians is:

$$c = 2 \operatorname{atan2}(\sqrt{a}, \sqrt{1-a}) \quad (4)$$

The surface distance $d(P_1, P_2)$ in meters is:

$$d(P_1, P_2) = R_E \cdot c \quad (5)$$

where $R_E = 6371000.0$ m is mean Earth radius. In `route_optimizer.cpp` and `drone.cpp`, distance calculations use this exact double-precision Haversine formulation.

B. Motion Model and Estimated Time of Arrival (ETA)

The aircraft's horizontal velocity $v(m)$ is determined by the active operational preset $m \in \{\text{Quick, Active, Inspect, Hover}\}$:

$$v(m) = \begin{cases} v_Q & m = \text{Quick} \\ v_A & m = \text{Active} \\ v_I & m = \text{Inspect} \\ v_H & m = \text{Hover} \end{cases} \quad (\text{m/s}) \quad (6)$$

where $v_H = 0$ is defined semantically as stationary hover.

Given current aircraft coordinates P_{drone} and target coordinates P_{dest} , remaining distance is $d = d(P_{\text{drone}}, P_{\text{dest}})$. For $v(m) > 0$, Estimated Time of Arrival (t_{ETA}) is formulated as:

$$t_{\text{ETA}} = \frac{d(P_{\text{drone}}, P_{\text{dest}})}{v(m)} \quad (\text{seconds}) \quad (7)$$

C. Discrete Battery Simulation Model

To exercise battery monitoring and failsafe behavior in the software prototype without requiring physical voltage/current telemetry, battery percentage $B_k \in [0, 100]$ evolves according to a discrete simulation rule:

$$B_{k+1} = \max(0.0, B_k - \alpha_m \cdot \Delta B) \quad (8)$$

where $\Delta B > 0$ is a step decrement constant and α_m is a mode-dependent prototype simulation coefficient. When the aircraft is landed ($v = 0, H = 0$), battery consumption is paused ($\Delta B = 0$). This discrete simulation rule exists solely to exercise software battery and failsafe behavior, and is not a physical electro-aerodynamic battery model.

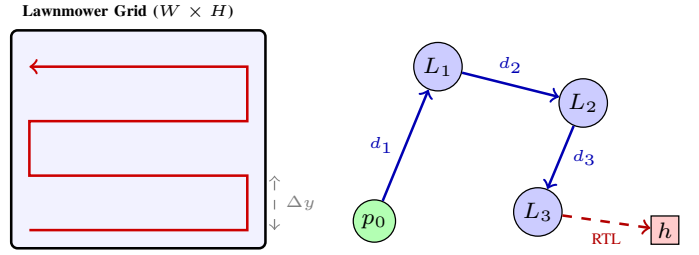


Fig. 3. Spatial geometry: Lawnmower sweep coverage pattern (left) and sequential greedy route progression with RTL return leg (right).

D. Lawnmower Search Grid Geometry

For a sector centered at (ϕ_c, λ_c) with sector width W along longitude and height H along latitude, conversion between meters and geographic coordinates uses local scale factors:

$$M_{\text{lat}} = 111320.0 \quad (\text{m / deg lat}) \quad (9)$$

$$M_{\text{lon}}(\phi_c) = 111320.0 \cdot \cos\left(\frac{\phi_c \cdot \pi}{180.0}\right) \quad (\text{m / deg lon}) \quad (10)$$

The southwest origin corner (ϕ_0, λ_0) of the grid is computed as:

$$\phi_0 = \phi_c - \frac{H/2.0}{M_{\text{lat}}}, \quad \lambda_0 = \lambda_c - \frac{W/2.0}{M_{\text{lon}}(\phi_c)} \quad (11)$$

Parallel sweep rows are separated by row spacing Δy .

VI. SEARCH ROUTE OPTIMIZATION

When operators enter multiple search locations, search order directly impacts total flight distance and mission duration. MANAR incorporates an independent greedy nearest-neighbor route optimizer. Fig. 3 visualizes the spatial sweep geometry and waypoint routing graph.

A. Problem Formulation

Let the operator define a set of n search locations:

$$L = \{L_1, L_2, \dots, L_n\}, \quad L_i = (\phi_i, \lambda_i) \quad (12)$$

Let $p_0 = (\phi_{\text{drone}}, \lambda_{\text{drone}})$ denote aircraft coordinates at plan lock time, and $h = (\phi_{\text{home}}, \lambda_{\text{home}})$ denote home base coordinates. Naive exhaustive permutation enumeration requires evaluating $n!$ waypoint orderings [4], which grows factorially and is impractical for onboard execution.

B. Greedy Nearest-Neighbor Formulation

MANAR implements an $O(n^2)$ greedy nearest-neighbor algorithm in `route_optimizer.cpp`. Starting from initial position p_0 , the algorithm iteratively selects the closest unvisited location.

Let U_k denote the set of unvisited locations at step k , with $U_0 = L$. For $k = 0, 1, \dots, n-1$:

$$p_{k+1} = \operatorname{argmin}_{x \in U_k} d(p_k, x) \quad (13)$$

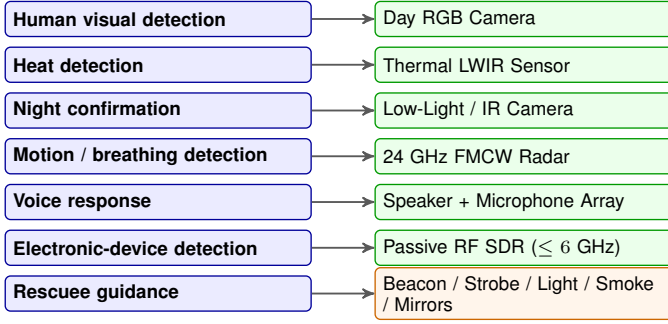


Fig. 4. Operational SAR capability to payload sensor and signaling subsystem mapping.

The selected location p_{k+1} is appended to the ordered route sequence, and the unvisited set is updated:

$$U_{k+1} = U_k \setminus \{p_{k+1}\} \quad (14)$$

Home location h is excluded from candidate selection iterations. The final leg $d(p_n, h)$ is included when calculating total planned transit distance d_{total} , without inserting h into the search sequence:

$$d_{\text{total}} = d(p_0, p_1) + \sum_{k=1}^{n-1} d(p_k, p_{k+1}) + d(p_n, h) \quad (15)$$

C. Combinatorial Complexity

The total number of candidate distance evaluations N_{eval} across n locations is:

$$N_{\text{eval}} = \sum_{k=0}^{n-1} (n-k) = \frac{n(n+1)}{2} = O(n^2) \quad (16)$$

For $n = 8$, nearest-neighbor selection requires 36 candidate distance evaluations ($8 + 7 + \dots + 1 = 36$), avoiding the $8! = 40320$ permutations of naive exhaustive search.

D. Operator Override Privilege

In tactical operations, operators may possess domain intelligence (e.g., wind drift, barrier terrain) that overrides geometric proximity. MANAR preserves operator authority:

$$\text{Route} = \begin{cases} \text{GreedyOpt}(p_0, L, h), & \text{if operator selects } Y \\ L, & \text{if operator selects } N \end{cases} \quad (17)$$

Disabling optimization (N) preserves exact input ordering ($p_k = L_k$). In both cases, `control.exe` freezes the plan (`searchplanlocked = true`) upon submission. In the current prototype, a locked plan remains immutable until the prototype reset path clears the lock.

VII. MULTISENSOR PAYLOAD ARCHITECTURE

MANAR incorporates a multi-modal payload suite engineered for victim search, identification, and rescue guidance. Fig. 4 illustrates the operational mapping between search requirements and sensor/signaling subsystems.

TABLE III
EMPIRICAL 26-VIDEO SAR BENCHMARK (YOLO11n vs. D-FINE-N)

Evaluation Metric	D-FINE-N	YOLO11n	Delta
Mean Latency (CPU)	50.5 ms	29.2 ms	-42.2%
Throughput (CPU FPS)	19.8 FPS	34.2 FPS	+72.7%
Model Parameters	≈ 4.0 M	≈ 2.6 M	-35.0%
Model File Size	14.55 MB	10.22 MB	-29.8%
Clutter False Positives	0 FP	0 FP	Parity
Concealed Target Detections	0 hits	44 hits	Superior
Multi-Rescuee Detections	531 hits	858 hits	+61.6%

A. Optical Subsystem and Ambient-Light Switching

To ensure continuous imaging across diurnal cycles, optical sensors transition dynamically based on ambient illuminance E_{lux} with hysteresis:

$$\text{Mode} = \begin{cases} \text{RGB} + \text{LWIR}, & E_{\text{lux}} > 150 \text{ lx} \\ \text{RGB} + \text{IR} + \text{LWIR}, & 50 \leq E_{\text{lux}} \leq 150 \text{ lx} \\ \text{IR} + \text{LWIR}, & E_{\text{lux}} < 50 \text{ lx} \end{cases} \quad (18)$$

B. Visual Candidate Detection & Empirical Benchmarking

MANAR implements a dedicated, high-throughput visual human candidate detector encapsulated in the C++ module `Yolo11nDetector` using **ONNX Runtime** (CPU/CUDA EP) and **OpenCV**. The module receives raw video frames and outputs instantaneous bounding-box candidate detections without embedding tracking heuristics inside the detector itself:

$$\mathbf{y}_k = \{\mathcal{B}_i, c_i, \text{class}_i\}_{i=1}^M, \quad c_i \geq c_{\text{th}} = 0.50 \quad (19)$$

Empirical SAR Dataset Evaluation: Both YOLO11n and the reference baseline D-FINE-N were evaluated across all 26 scenario evaluation videos in MANAR’s training dataset spanning RGB, low-light IR, and thermal streams under identical parameters (640×640 input, $c_{\text{th}} = 0.50$, $\text{IoU}_{\text{NMS}} = 0.45$). As summarized in Table III:

- 1) **Inference Speed:** YOLO11n achieved a 29.2 ms mean latency per frame, running $\approx 1.73\times$ faster than D-FINE-N (50.5 ms), providing real-time 34 FPS throughput on embedded CPU cores.
- 2) **Negative Rejection:** Both architectures produced zero false positives across all clean-negative and environmental clutter scenes in all three spectral bands.
- 3) **Concealed & Multiple Targets:** YOLO11n demonstrated superior detection density on partially concealed subjects (44 vs. 0 detections) and multi-rescuee formations (858 vs. 531 detections).

Consequently, YOLO11n is locked as MANAR’s V1 visual detector, while D-FINE-N is retained as the reference evaluation baseline.

C. Acoustic Subsystem and Call-and-Listen Cycle

The acoustic payload incorporates a 4–8 digital MEMS microphone array mounted on vibration-isolated damping brackets. Rotor harmonics are actively suppressed using real-time

motor RPM references across the 200 Hz–8 kHz processing band.

During hover hold, the system executes a structured 35.4 s call-and-listen cycle:

- 1) **Acoustic Alert (8.4 s):** Standard SOS buzzer sequence (3× short 0.4 s, 3× long 1.2 s, 3× short 0.4 s).
- 2) **Voice Prompt (12.0 s):** Spoken instructions in English and Arabic: “*Rescue drone. Shout, whistle, clap, or hit an object now*”.
- 3) **Silent Listening (15.0 s):** Emitters are silenced while the array records ambient signals, evaluating Beamforming Direction of Arrival (DoA) and classifying vocal distress events ($\geq 70\%$ confidence).

D. Passive RF Sensing and Two-Level Attention

Passive RF monitoring scans sub-6 GHz bands (cellular uplinks 600 MHz–2.7 GHz, sub-6 5G, Wi-Fi 2.4/5 GHz, and Bluetooth 2.4 GHz; > 6 GHz excluded) using dual directional antennas:

- **Level 1 Attention (CFAR Detection):** Activates when at least three qualifying samples exceed the adaptive CFAR threshold within an observation window ($N_{\text{CFAR}+} \geq 3$).
- **Level 2 Attention (Temporal Trend Tracking):** Engages *Inspect* mode with elevated sampling. The local trend $\Delta S_k = S_k - \frac{1}{m} \sum_{i=k-m}^{k-1} S_i$ is classified as HOT ($\Delta S_k > \epsilon$), STABLE ($|\Delta S_k| \leq \epsilon$), or COLD ($\Delta S_k < -\epsilon$).

E. FMCW Radar Ranging Characteristics

Operating at a 24 GHz center frequency with sweep bandwidth $B \approx 250$ MHz, the theoretical range resolution ΔR is:

$$\Delta R \approx \frac{c}{2B} \approx \frac{3 \times 10^8 \text{ m/s}}{2 \times 250 \times 10^6 \text{ Hz}} \approx 0.60 \text{ m} \quad (20)$$

F. Planned Mamba Perception & Fusion Pipeline

Fig. 5 illustrates the planned temporal perception architecture.

Synchronized feature vectors $\mathbf{x}_t \in \mathbb{R}^D$ sampled at 10 Hz across 20–50 time steps (2–5 s window) are evaluated by a Mamba State Space Model (SSM) [2]. Mamba acts strictly as an alert escalation filter (Reject candidate, Continue verification, or Alert operator). Final classification (False positive, Manual continued inspection, or Probable rescuee) is reserved exclusively for the human operator.

G. Environmental Complementarity and Rescue Guidance

Table IV summarizes the operational robustness of the sensing modalities across adverse environmental conditions.

Rescue Guidance Systems: Active guidance integrates a 360° amber flashing beacon, a 2 Hz directional white strobe, a downward spotlight for night inspection, and a pyrotechnic smoke marker for wind/location marking. Supplementary passive daylight signaling is provided by three fixed heliograph mirrors mounted at 40°, 60°, and 75° from vertical, producing ground reflections at nominal standoff distance $R = H \tan \theta$.

TABLE IV
SENSOR MODALITY ROBUSTNESS IN ADVERSE ENVIRONMENTS

Modality	Day	Night	Rain	Dust	Smoke
Thermal LWIR	High	High	Mod	Mod	High
Day RGB	High	Low	Mod	Low	Low
Low-Light IR	Mod	High	Mod	Low	Mod
24 GHz FMCW	High	High	High	High	High
Acoustic Array	High	High	Low	Mod	Mod
Passive RF	High	High	High	High	High

VIII. ADAPTIVE TELEMETRY AND RADIO POWER MODEL

To preserve aircraft battery endurance while maintaining operational situation awareness, MANAR defines an adaptive telemetry transmission model. This section formalizes the radio power consumption model as a function of stream frequency and establishes a multi-tiered transmission rate policy.

A. Analytical Radio Telemetry Power Model

Radio frequency (RF) telemetry power consumption scales with transmission duty cycle. Let P_{TX} denote transmitter active power, P_{RX} denote receiver baseline power, and R denote link channel data rate in bits per second (bps).

For a telemetry stream i with message payload size B_i (bytes) transmitted at frequency f_i (Hz), the incremental electrical radio power ΔP_i consumed over the idle/receive baseline is formulated as:

$$\Delta P_i = (P_{\text{TX}} - P_{\text{RX}}) \cdot \left(\frac{8 \cdot B_i \cdot f_i}{R} \right) = \beta \cdot B_i \cdot f_i \quad (21)$$

where β is the radio transmission slope coefficient:

$$\beta = \frac{(P_{\text{TX}} - P_{\text{RX}}) \cdot 8}{R} \left(\frac{\text{W}}{\text{byte} \cdot \text{Hz}} \right) \quad (22)$$

For a reference UHF/cellular telemetry link with $P_{\text{TX}} = 4.80$ W, $P_{\text{RX}} = 0.36$ W, and channel rate $R = 64\,000$ bit/s, the slope coefficient evaluates to $\beta \approx 5.55 \times 10^{-4}$ W/(byte · Hz).

Fig. 6 illustrates the incremental radio power across transmission frequencies from 0 to 10 Hz for fourteen distinct MANAR telemetry message categories.

B. Adaptive Transmission Rate Policy

To minimize RF power overhead during transit while providing high-bandwidth telemetry during active victim localization, MANAR implements a dual-regime rate policy. Table V defines the transmission rates for all telemetry streams under normal search vs. proximity/critical operations.

Under normal cruise and transit search phases, routine navigation streams (position, speed, altitude) update at 2 Hz, while slow-varying state like battery percentage is throttled to 0.2 Hz (5 s intervals), and discrete states transmit strictly upon change. When candidate evidence triggers a single-sensor or multisensor confidence event, the system dynamically escalates to the proximity rate regime (5–10 Hz), allocating RF bandwidth and power specifically when localization fidelity is paramount.

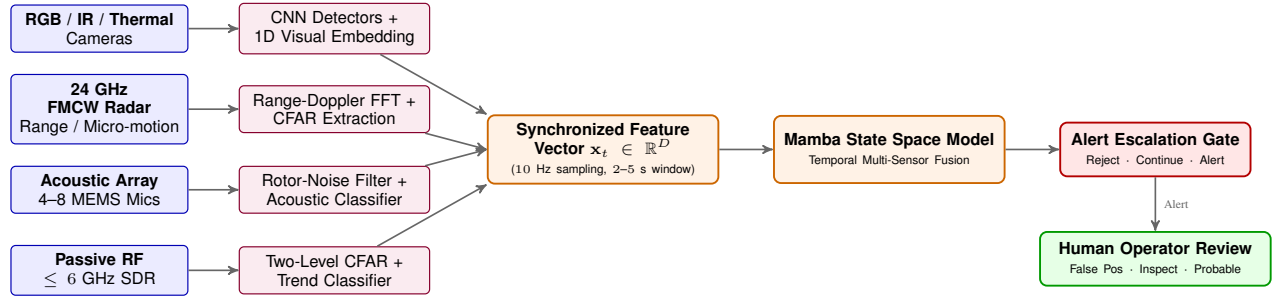


Fig. 5. Planned multisensor perception and Mamba hover verification pipeline (Planned Future Architecture).

TABLE V
MANAR ADAPTIVE TELEMETRY RATE POLICY AND OPERATIONAL RATIONALE

Tier	Telemetry Stream	Payload (B_i)	Normal Rate	Proximity / Critical Rate	Operational Rationale
●	Heartbeat	24 B	2.0 Hz	2.0 Hz	Low-overhead link-health and watchdog awareness.
●	Position	28 B	2.0 Hz	5.0 Hz	Primary navigation telemetry; modest radio power cost.
●	Speed	24 B	2.0 Hz	5.0 Hz	Low power footprint; useful for live kinematic context.
●	Altitude	24 B	2.0 Hz	5.0 Hz	Critical for terrain clearance and vertical safety.
●	Heading	22 B	1.0 Hz	5.0 Hz	Low marginal cost; 1 Hz baseline sufficient in transit.
●	Battery Percentage	22 B	0.2 Hz	0.5 Hz	5-second interval sufficient under normal discharge.
●	Flight Mode	21 B	On change	On change	Event-driven update; redundant transmissions avoided.
●	Mission State	21 B	On change	On change	State transition telemetry published on event trigger.
●	Component Status	24 B	On change / fault	Active components only	Payload status updates sent on state modification.
●	Sensor Measurements	52 B	Local only	5.0–10.0 Hz	High-rate raw sensor data streamed only in proximity.
●	Single-Sensor Confidence	28 B	Immediate event	Stream relevant sensor	Triggered immediately upon validated detection event.
●	Multisensor Confidence	32 B	Immediate event	Trigger proximity mode	High-priority event; escalates system telemetry tier.
●	Fault / Failsafe / Low Batt	28 B	Immediate event	Repeat while active	Emergency telemetry; bypasses normal bandwidth caps.
●	Diagnostics / Logs / Config	84 B	On-demand	On-demand	High payload footprint; retrieved only on operator query.

IX. SAFETY, AUTHORIZATION, AND FAILSAFES

Safety rules and deterministic monitoring logic are enforced directly within the C++ control core.

A. Supervised Operator Authorization Rules

To maintain operational integrity in high-stakes SAR environments, MANAR enforces six core supervisory rules:

- 1) **Human Confirmation Authority:** Machine learning models gate candidate alerts, but only the human operator is authorized to assign *Probable Rescuee* status.
- 2) **Multi-Sensor Consensus:** No single sensor modality dominates victim confirmation; multi-modal evidence correlation is required.
- 3) **Supporting Modality Principle:** Thermal LWIR and passive RF detections are treated strictly as supporting

cues, avoiding false alarms from heated terrain or abandoned devices.

- 4) **Guidance Authorization:** Active signaling devices (beacon, strobe, spotlight, smoke) require explicit operator authorization before activation.
- 5) **Restricted Smoke Deployment:** Smoke marker release is strictly prohibited near fire-prone vegetation, fuel storage, crowds, or adverse wind conditions.
- 6) **Safety Precedence:** Flight safety, airframe structural limits, icing risk, and minimum return energy strictly override search or detection confidence.

B. Communications Resilience and Link-Loss Logging

Low-bandwidth flight telemetry (2–10 Hz) is transmitted across the primary encrypted command-and-control channel independently of high-bandwidth video streams (*SYSS-R-03*).

Incremental Telemetry Power vs. Transmission Frequency

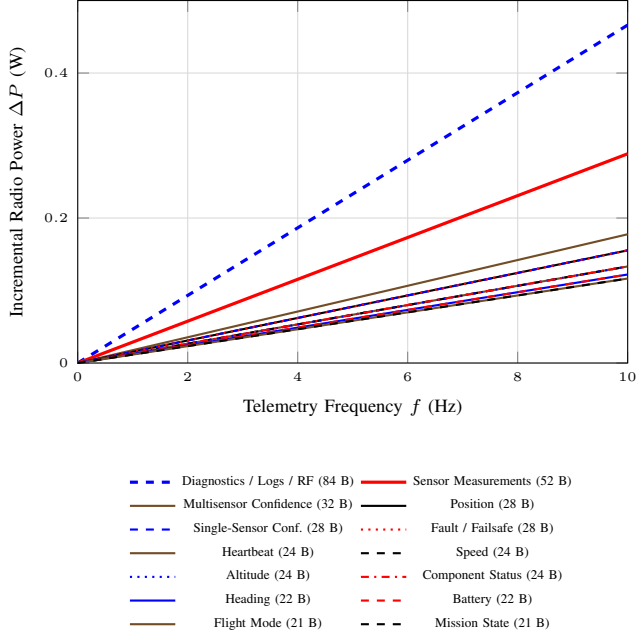


Fig. 6. Analytical estimate of incremental radio power for individual MANAR telemetry message types across transmission frequencies ($P_{TX} = 4.8$ W, $P_{RX} = 0.36$ W, $R = 64$ kbps).

During total external communication loss, the control core logs all telemetry, flight modes, and sensor health data to onboard non-volatile storage while executing link-loss contingencies.

C. Battery Failsafe Hierarchy

The control core monitors battery percentage B_k on every loop iteration:

$$\text{Action}(B) = \begin{cases} \text{Emergency Land,} & B_k \leq B_{\text{land}} \\ \text{Emergency RTL,} & B_k \leq B_{\text{emergency_rtl}} \\ \text{Recommend RTL,} & B_k \leq B_{\text{rtl_warn}} \\ \text{Battery Warning,} & B_k \leq B_{\text{warn}} \\ \text{Normal Operation,} & \text{otherwise} \end{cases} \quad (23)$$

When $B_k \leq B_{\text{emergency_rtl}}$, `activateRTL()` automatically preempts active search and commands Return-to-Launch navigation to home base.

D. Software Command Clamps

All velocity and altitude command arguments are validated against software clamps prior to execution:

$$v_{\text{cmd}} \leq v_{\text{max}} \quad (24)$$

$$H_{\text{cmd}} \leq H_{\text{max}} \quad (25)$$

Commands exceeding these limits are clamped, logging WARN events to `runtime/logs.txt`.

TABLE VI
MANAR C++ MODULE ARCHITECTURE

Module	Type	Core Responsibilities
<code>control.cpp</code>	App	Control loop, JSON command dispatch, runtime state serialization.
<code>terminal.cpp</code>	App	Coordinate entry, DMS parsing, optimization toggle, operator commands.
<code>setup.cpp</code>	App	Configuration slot selection, parameter template initialization.
<code>mission.cpp</code>	Core	Lawnmower sweep progression, search plan locking, RTL preemption.
<code>drone.cpp</code>	Core	Geodetic kinematics, distance-to-home, discrete battery simulation.
<code>flight.cpp</code>	Core	Velocity/altitude clamps, flight modes (Quick, Active, Inspect, Hover).
<code>route_optimizer.cpp</code>	Core	Haversine geodetics, $O(n^2)$ greedy nearest-neighbor solver.
<code>components.cpp</code>	Core	State control for 11 controllable payload categories.
<code>shared.cpp</code>	Core	Timestamped structured plaintext event logging.
<code>yolo11n_detector.cpp</code>	CV	Reusable YOLO11n ONNX Runtime inference wrapper, letterbox preprocessing, NMS.
<code>yolo11n_video.cpp</code>	CV App	Video file evaluation harness with annotated frame output.
<code>yolo11n_live.cpp</code>	CV App	Live camera / RTSP stream detection test application.
<code>dfine_video.cpp</code>	CV App	D-FINE-N reference baseline video evaluation harness.

X. SOFTWARE ARCHITECTURE

MANAR is implemented as a modular C++ system structured into standalone applications and decoupled domain subsystems.

A. C++ Core & Perception Domain Modules

Table VI summarizes the architectural responsibilities of the core C++ source modules across control and perception subsystems.

B. Separation of Software Layers

MANAR enforces architectural separation across decoupled operational layers:

- **C++ Control Core (`core/`):** Executes deterministic mission state machine, kinematic flight bounds, and greedy route optimization.
- **Visual Perception Library (`computer_vision/`):** High-performance ONNX Runtime C++ detector module producing instantaneous candidate detections.
- **Operator Interface (`React/TypeScript`):** Selected web-based operator terminal communicating via WebSocket + JSON IPC protocol.
- **Temporal Fusion Engine (`Mamba SSM`):** Planned multi-modal evidence association and candidate alert escalation filter.

XI. FUNCTIONAL PROTOTYPE VERIFICATION

The prototype implementation of MANAR has been built and functionally verified under Windows 11 using GCC 16.1.0 (g++).

A. Build System

Core control and perception components are compiled into standalone executables and shared/static libraries:

```
# 1. C++ Control Core Applications
cd core && mkdir build
g++ -std=c++17 apps/control.cpp system/shared.cpp
system/flight.cpp system/components.cpp
system/drone.cpp system/mission.cpp
system/route_optimizer.cpp
-I. -Ithird_party -o build/control.exe
g++ -std=c++17 apps/terminal.cpp -I. -Ithird_party
-o build/terminal.exe
g++ -std=c++17 apps/setup.cpp -I. -Ithird_party
-o build/setup.exe

# 2. Computer Vision Pipeline (CMake / ONNX Runtime)
cd ../computer_vision && mkdir build && cd build
cmake .. -DONNXRUNTIME_ROOT_DIR=<path_to_ort>
cmake --build . --config Release
```

Listing 1. MANAR C++ core and computer vision compilation commands.

B. Functional Verification Results

Functional testing confirmed the following behaviors:

- 1) **Route Optimization vs. Manual Order:** For three search coordinates (L_1, L_2, L_3) entered in non-optimal geometric order relative to start position p_0 , enabling optimization (Y) produced a shorter planned transit distance than the entered manual ordering ($d_{\text{greedy}} < d_{\text{manual}}$), while disabling optimization (N) preserved exact entry order.
- 2) **Search Plan Locking:** Submitting locations locked `search_locations.json`. Subsequent `SET_SEARCH_LOCATIONS` commands were rejected with WARN logs until the prototype reset path cleared the lock.
- 3) **Start Rejection on Empty Plan:** Attempting `START_MISSION` without search locations resulted in immediate rejection by both terminal and control core.
- 4) **RTL Precedence:** Triggering RTL during multi-sector search cleared search waypoints, reset `search_locations.json` to a single home destination, launched the aircraft if grounded, and initiated return navigation using the configured flight mode.
- 5) **Visual Inference Verification:** The C++ `yolo11n_video` and `yolo11n_live` applications were verified against synthetic and empirical SAR video feeds, confirming correct candidate bounding-box extraction, confidence thresholding, letterbox rescaling, and ONNX Runtime session persistence without memory leakage.

TABLE VII
PROTOTYPE VERIFICATION TEST PARAMETERS

Symbol	Parameter	Value
v_Q	Quick velocity	15.0 m/s
v_A	Active velocity	8.0 m/s
v_I	Inspect velocity	2.0 m/s
v_H	Hover velocity	0.0 m/s
H_{launch}	Launch altitude	10.0 m
H_{max}	Altitude ceiling	2000 m
v_{max}	Speed ceiling	15.0 m/s
$W \times H$	Sector dimensions	100×100 m
Δy	Row spacing	20.0 m
R_{reach}	Waypoint arrival radius	5.0 m
R_{home}	Home arrival radius	100 m
B_{warn}	Warning threshold	30%
$B_{\text{rtl_w}}$	Recommend RTL	25%
B_{emg}	Emergency RTL	15%
B_{land}	Emergency Land	5%
α_m	Battery sim. coefficients	5, 3, 2, 1

C. Prototype Test Configuration

Table VII lists the configuration parameters used during functional verification. These values are implementation-specific test inputs and do not constitute finalized MANAR operational limits.

XII. LIMITATIONS AND FUTURE WORK

While MANAR establishes a deterministic C++ baseline and comprehensive architectural framework, several operational limitations warrant ongoing development:

A. Environmental and Weather Boundaries

MANAR does not claim all-weather operational capability:

- **Precipitation and Icing:** Heavy rain attenuates optical/acoustic signals, while dense wet snow and hail introduce propeller icing, airframe structural damage, and severe propulsion risk.
- **Atmospheric Obscurants:** Dense fog degrades visual/IR imaging; dust and sandstorms introduce optical backscatter and motor bearing abrasion.
- **Thermal and RF Clutter:** Solar-heated desert rocks, hot vehicle engines, and active fires create thermal false alarms. Canyons, metallic structures, and urban rubble induce multipath and bearing errors in passive RF localization.
- **Radar and Acoustic Contamination:** Wind-blown vegetation can mimic human micro-motion on FMCW Doppler maps, while airframe vibration and rotor noise can mask faint acoustic distress calls.

B. Active Search Endurance Constraints

The 1.0 hr active search endurance target (SYS-G-02) depends strongly on aggregate payload mass, battery specific energy, ambient operating temperature, search velocity, hover verification frequency, onboard compute draw, and active guidance usage. Full empirical validation on the integrated airframe remains planned future work.

C. Algorithmic Bounds and Future Work

The greedy route optimizer executes in $O(n^2)$ time but does not guarantee global optimality; future iterations may explore local 2-opt search or exact Mixed-Integer Linear Programming (MILP). With the raw visual detector (YOLO11n) established, immediate future work focuses on multi-frame candidate persistence (ByteTrack / Kalman filtering), implementing the React/TypeScript GUI over WebSocket, validating the Mamba SSM on field datasets, integrating PX4 Software-in-the-Loop (SITL) simulation, and conducting physical flight trials.

XIII. CONCLUSION

This paper presented MANAR (Multi-sensor Autonomous Navigation and Aerial Rescue), a supervised-autonomy SAR system architecture built upon a deterministic C++ control core, Haversine geographic navigation, an $O(n^2)$ greedy search route optimizer with operator override privilege, and an ONNX Runtime C++ visual candidate detector (YOLO11n). By maintaining a strict distinction between the implemented functional C++ prototype, selected software state ownership architecture, and planned perception extensions, the paper establishes a grounded technical baseline. Functional prototype verification confirmed deterministic control behavior, search-plan locking, automatic Return-to-Launch (RTL) navigation preemption, and real-time visual candidate inference (29.2 ms latency with zero false positives on clutter). MANAR establishes a foundation for dependable, human-supervised aerial search-and-rescue robotics.

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